

Multi-Modal Deepfake Detection with Frozen Self-Supervised Backbones, Cross-Region Attention, and Adaptive Gated Fusion: A Systematic Cross-Dataset Generalization Study

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Abstract—Deepfake detectors routinely report near-perfect accuracy on their own benchmark’s held-out test split, yet the practical value of such a system depends on whether its performance survives exposure to previously unseen generation methods. We present a two-stage, dual-modality deepfake detection system that combines frozen self-supervised visual and audio backbones (DINOv2 ViT-S/14 and Whisper-tiny), a four-region (face/eyes/lips/jaw) cross-region attention module, per-modality bidirectional LSTMs, a two-layer fusion Transformer, and a Gated Multimodal Unit (GMU) for adaptive visual-audio weighting, together with a low-rank adaptation (LoRA) fine-tuning stage and an auxiliary multi-class forgery-type head. Trained on a pool of 44,561 videos from FakeAVCeleb, PolyGlottFake, FaceForensics++, and Celeb-DF-v2 (the latter was previously mislabeled as FaceForensics++ in this pipeline; both this and a separate identity-leakage bug in the train/val/test splitting logic were found and fixed as part of this work, Section IV-A), the system reaches AUC 0.998 and >98% accuracy in-distribution – verified by retraining *both* of our two candidate Stage B recipes against the corrected, genuinely identity-disjoint split and re-selecting canonical by validation AUC exactly as our own methodology requires (Section IV-A), rather than reporting an unverified number: the selected recipe is unchanged from before the fix, in-distribution AUC was essentially unaffected by the leakage bug (0.9962→0.9984 for this recipe), and DFDC cross-dataset AUC under the corrected split (0.7091) is in line with our original finding (0.71). Critically, we reserve the DeepFake Detection Challenge (DFDC) dataset entirely for out-of-distribution evaluation, excluded from every stage of training, validation, and threshold selection. Critically, DFDC is also excluded from *configuration selection*: the deployed configuration is chosen using validation AUC alone, and DFDC results for all five candidate Stage B configurations are then reported as a single blind post-hoc evaluation, rather than retrospectively reporting whichever configuration happened to score best on DFDC. On this genuinely unseen distribution, AUC ranges from 0.65–0.71 across five Stage B configurations and one Stage A prerequisite baseline (frozen-backbone baseline, LoRA fine-tuning, data augmentation, weight decay, Self-Blended Images synthetic data, and an SRM frequency-domain branch), none of which closes more than a fraction of the gap. We complement the ablation with quantitative error analysis, a modified Grad-CAM/attention-rollout explainability pipeline, and a causal boundary-cue masking ex-

periment; the latter contradicts the boundary-reliance hypothesis the Grad-CAM visualization initially suggested, illustrating that attention-map-based diagnosis requires quantitative verification rather than acceptance at face value. We report this observed generalization limit as a primary finding rather than obscuring it behind in-distribution numbers, and argue that closing it requires training-data diversity across generation-method families rather than further architectural or regularization tuning on the same three datasets.

Index Terms—deepfake detection, multi-modal learning, self-supervised learning, DINOv2, Whisper, low-rank adaptation, gated multimodal unit, cross-dataset generalization, explainable AI, Grad-CAM

I. INTRODUCTION

The proliferation of generative adversarial networks (GANs), autoencoder face-swap pipelines, and diffusion-based synthesis tools has made photorealistic facial forgery – *deep-fakes* – inexpensive to produce and increasingly difficult to detect by eye [?]. The resulting risks to information integrity, identity security, and public trust in audiovisual media have driven a large body of detector research, almost all of which is evaluated on the same handful of public benchmarks [?], [?], [?].

A persistent, under-reported problem in this literature is that *in-distribution* accuracy – performance on a held-out split of the *same* dataset(s) used for training – is a poor proxy for real-world usefulness. Detectors learn to recognize the specific compression artifacts, generator fingerprints, and post-processing signatures of the generation pipelines present in their training data, and that signal frequently fails to transfer to a video produced by a different, unseen generation method [?], [?], [?]. A detector that reports 99% accuracy on its own benchmark but silently degrades to substantially degraded performance on a forgery technique it has not seen cannot be considered solved in a deployment setting.

This paper makes two contributions. First, we describe a complete multi-modal deepfake detection architecture built

around frozen self-supervised visual (DINOv2 [?]) and audio (Whisper [?]) backbones, fine-tuned efficiently with low-rank adaptation (LoRA) [?], and fused adaptively across modalities with a Gated Multimodal Unit (GMU) [?]. The architecture additionally uses region-level cross-attention across four facial regions, bidirectional LSTM temporal modeling, an auxiliary multi-class manipulation-type head, and an explainability pipeline that combines a DINOv2-specific adaptation of Grad-CAM [?] with Transformer attention rollout [?].

Second, and more centrally, we treat cross-dataset generalization as a first-class experimental question rather than an afterthought. We reserve the DFDC dataset [?] in its entirety – never touched during training, validation, decision-threshold selection, or, critically, *configuration selection* (Section V-C) – as a probe of genuine out-of-distribution performance, and report six independent attempts to close the resulting in-distribution/cross-dataset gap: frozen-backbone extraction, LoRA fine-tuning, data augmentation, weight decay regularization, Self-Blended Images (SBI) synthetic data [?], and a hand-engineered SRM frequency-domain branch [?]. We report the outcome of every attempt, including interventions that yielded no improvement or reduced performance, together with a quantitative error analysis and an explainability-driven diagnosis of *why* the gap persists. We believe this transparency – reporting the observed 0.71 AUC generalization limit rather than only the 0.996 in-distribution headline number – is itself a useful empirical contribution, consistent with the community’s growing recognition that cross-dataset evaluation, not in-dataset evaluation, is the meaningful test of a deepfake detector [?], [?], [?].

The remainder of this paper is organized as follows. Section II reviews related work. Section III describes the proposed architecture. Section IV details the datasets, splitting protocol, and training procedure. Section V presents in-distribution and cross-dataset results, the six-attempt generalization ablation, and the diagnostic analysis. Section VI discusses implications and limitations. Section VII concludes.

II. RELATED WORK

A. Spatial and Frequency-Domain Detectors

Early deepfake detectors treated the problem as binary image classification, using CNN backbones such as Xception [?] or compact purpose-built networks such as MesoNet [?] to pick up mesoscopic blending artifacts. FaceForensics++ [?] established the standard evaluation protocol for this line of work. Frequency-domain approaches complement spatial CNNs by exposing GAN up-sampling artifacts that are less visible in the spatial domain; SRM-style high-pass residual filters, originally developed for steganalysis [?], are a common hand-engineered choice for this purpose, and learned frequency-aware branches such as F3-Net [?] extend the idea to end-to-end training. A related line of work targets the generic blending boundary left by any face-compositing operation rather than a specific generator’s fingerprint, exemplified by Face X-Ray [?]; our Self-Blended Images attempt (Section V-C) is built on this boundary-artifact framing, and our Grad-CAM spot-check

(Section V-D) initially appeared to connect to it as well, though a subsequent causal ablation (also Section V-D) did not bear that reading out.

B. Temporal and Multi-Modal Detectors

Because many forgeries are frame-consistent but temporally or cross-modally inconsistent, later work incorporates recurrent or attention-based temporal modeling (LSTMs [?], [?], Transformers [?]) and audio-visual synchronization cues, as in LipForensics [?] and FTCN [?], both of which report substantially better cross-manipulation generalization than purely spatial detectors. The DFDC [?] and FakeAVCeleb [?] datasets were released specifically to support multi-modal and cross-manipulation-method evaluation; PolyGlotFake extends this to multilingual audio-driven forgeries [?]. Gated Multimodal Units [?] provide a principled, learned mechanism for weighting unreliable or missing modalities, which we adopt for visual-audio fusion.

C. Self-Supervised Backbones and Parameter-Efficient Fine-Tuning

Self-supervised vision backbones such as DINOv2 [?] and speech backbones such as Whisper [?] are trained on web-scale unlabeled or weakly-labeled data and produce general-purpose representations that transfer well across downstream tasks. Because full fine-tuning of such backbones on a comparatively small forgery dataset risks catastrophic overfitting to dataset-specific artifacts, we use Low-Rank Adaptation (LoRA) [?], which freezes the backbone and injects small trainable low-rank update matrices into a subset of layers.

D. Synthetic Data for Generalization

Self-Blended Images (SBI) [?] synthesize pseudo-forgeries from real images alone, by blending a warped, color-jittered copy of a face back onto itself, with the goal of teaching a generic “blending-boundary” cue that is not tied to any specific generator’s fingerprint. We include SBI as one of our six generalization-improvement attempts.

E. Explainability

Grad-CAM [?] was designed for convolutional architectures and does not transfer directly to Vision Transformers that pool only a CLS token, since patch tokens frequently have no differentiable path to the classification logit. Attention rollout [?] addresses a related problem for Transformer stacks by recursively composing per-layer attention matrices (with an identity term for the residual connection) into a single input-attribution map. We combine an architecture-specific Grad-CAM adaptation for DINOv2 with attention rollout over our fusion Transformer, described in Section III-I.

III. PROPOSED ARCHITECTURE

Figure ?? summarizes the pipeline. The system consumes a video and produces (i) a real/fake probability, (ii) an auxiliary manipulation-type distribution, and (iii) a GMU gate value indicating the relative contribution of the visual and audio streams.

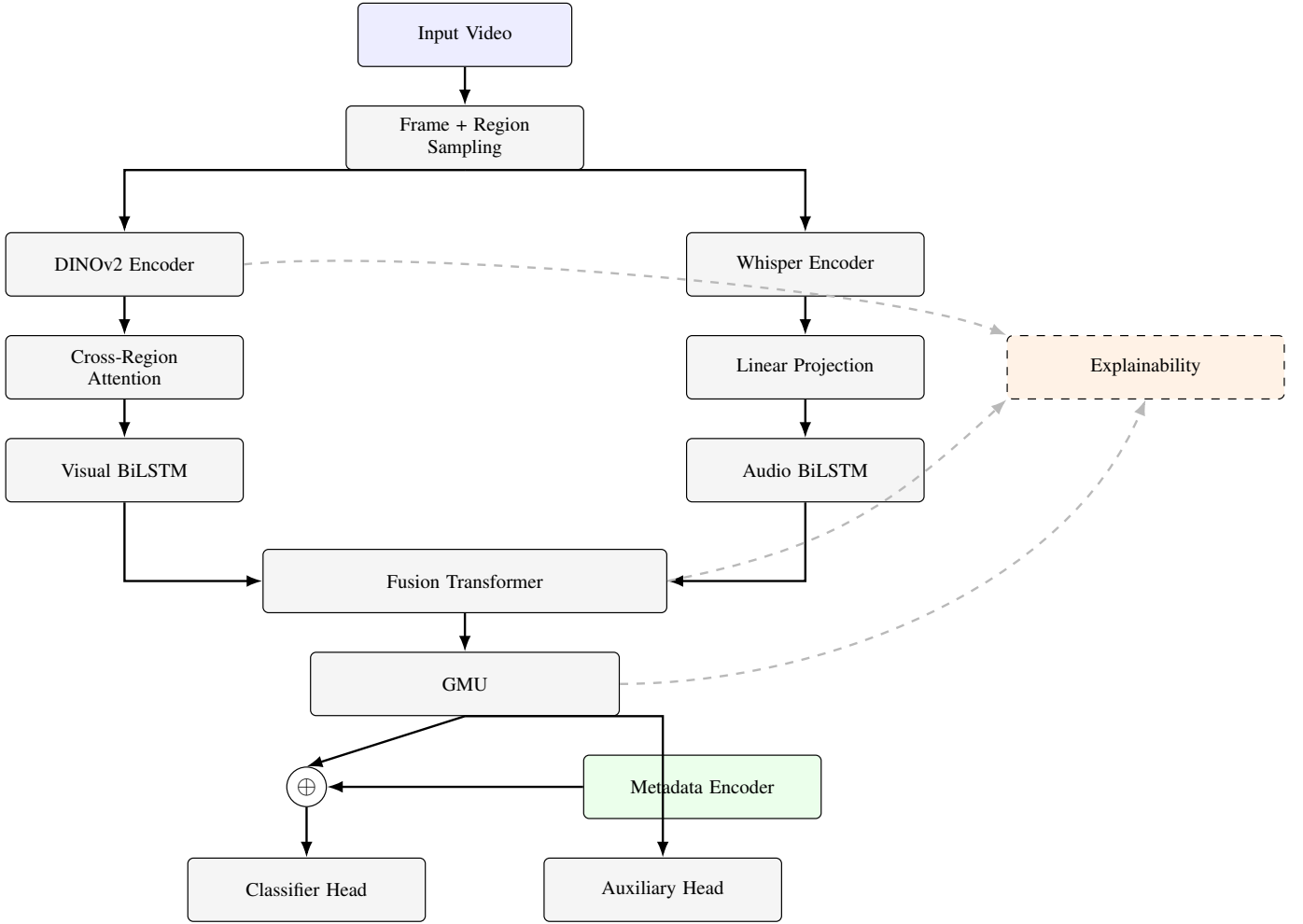


Fig. 1. End-to-end architecture (implementation detail omitted from the diagram for readability is given here). Frame + Region Sampling: 8 uniformly-sampled frames, 4 region crops per frame (face / eyes / lips / jaw). DINOv2 / Whisper Encoders: DINOv2 ViT-S/14 and Whisper-tiny, both frozen in Stage A and partially re-opened via LoRA adapters in Stage B (the same backbone stack is used at both stages; Stage A caches frozen features, Stage B trains end-to-end on raw crops). Cross-Region Attention: a region-projection layer followed by multi-head self-attention across the 4 region tokens per frame (Section III-D). Fusion Transformer: 2 layers, 4 heads, with learned modality embeddings distinguishing visual from audio tokens (Section III-F). GMU: the Gated Multimodal Unit that adaptively weights the pooled visual and audio streams (Section III-F). Metadata Encoder: a small MLP over the 34-dimensional ffprobe -derived feature vector (Section III-G). The $\oplus$ node denotes concatenation of the GMU’s fused representation with the metadata embedding, which feeds only the Classifier Head (real/fake logit); the Auxiliary Head (9-way manipulation-type classification) reads the fused representation directly, without metadata. Dashed arrows/box indicate the explainability pathway (Section III-I) – Grad-CAM over DINOv2’s CLS-to-patch attention, attention rollout over the fusion Transformer, and the GMU gate value – which consumes intermediate model state but does not feed back into the classifier.

A. Frame and Region Sampling

Eight frames per video are selected by uniform temporal index sampling,

$$t_i = \left\lfloor \frac{iN}{8} \right\rfloor, \quad i \in \{0, 1, \dots, 7\}, \quad (1)$$

given N total frames, so that the sample spans the full video rather than only its opening seconds. The frame count was set, together with the batch-size and gradient-accumulation choices of Section IV-B, to fit within the 8 GB GPU memory budget available for this project; we did not run a systematic ablation over the frame count, so denser sampling (e.g., 16 or 32 frames) remains an open direction rather than a value we can claim is optimal. Faces are localized with MediaPipe Face Landmarker [?] (468 landmarks, minimum detection

confidence 0.4), with MTCNN [?] as a coarse-bounding-box fallback and the full frame as a last resort when neither detector fires. From the resulting face bounding box (x_1, y_1, x_2, y_2) with height $f_h = y_2 - y_1$, four regions are cropped and resized to 224×224 : *face* (the full box), *eyes* (top 40% of box height), *lips* (bottom 40%), and *jaw* (from 75% down to 15% below the box, chosen because face-swap blending seams concentrate around the jawline-to-neck boundary). These four regions, rather than e.g. the forehead or cheeks, were chosen because prior forgery-detection work consistently localizes artifacts to facial boundaries and blending seams [?], the periocular region (blink-rate and gaze anomalies), and the lips (audio-visual synchronization [?]).

B. Visual Feature Extraction: DINOv2

Each of the four 224×224 region crops, for each of the eight frames, is encoded independently by DINOv2 ViT-S/14 [?], a self-supervised Vision Transformer [?] trained with a student-teacher distillation objective on large-scale curated image data. We retain only the 384-dimensional CLS token per region per frame, giving a visual feature tensor of shape $(8, 4, 384)$ per video. DINOv2 is chosen over a supervised ImageNet backbone because its self-supervised objective encourages representations of generic visual structure rather than class-discriminative shortcuts, which is desirable when the discriminative task at hand (forgery detection) is not the task the backbone was trained for. The ViT-S/14 variant specifically was selected, over the larger ViT-B/14 or ViT-L/14 variants, to fit the project’s 8 GB GPU memory budget (Section IV-B); we did not run a comparative ablation against the larger DINOv2 variants, so we cannot claim ViT-S/14 is optimal, only that it is the variant this budget supported.

C. Audio Feature Extraction: Whisper

The audio track is encoded with the Whisper-tiny encoder [?], a Transformer speech encoder trained via large-scale weakly-supervised multitask learning. Whisper-tiny (39M parameters), the smallest variant in the Whisper family, was likewise selected to fit the same 8 GB memory budget as the visual backbone; larger Whisper variants (base, small, and above) were not evaluated in this project. Whisper’s encoder emits a fixed 1500-step sequence (from its 30-second-padded input convention), which is adaptively average-pooled (`adaptive_avg_pool1d`) down to 8 temporal steps to align exactly with the 8 visual frames – a design choice that lets the fusion Transformer (Section III-F) attend across modalities token-for-token without any additional interpolation or asynchronous token-matching step, at the cost of discarding whatever finer-grained temporal audio structure exists within each of the 8 pooled windows. This produces an audio feature tensor of shape $(8, 384)$.

D. Cross-Region Attention

The four per-frame region embeddings are first linearly projected from 384 to the model’s working dimension $d = 256$. A learned per-region embedding is added (so the attention layer can distinguish face/eyes/lips/jaw identity), and a standard multi-head self-attention layer (4 heads) lets the four regions attend to each other within each frame:

$$\mathbf{a}_t = \text{LayerNorm}(\text{MHA}(\mathbf{r}_t, \mathbf{r}_t, \mathbf{r}_t) + \mathbf{r}_t), \quad (2)$$

where $\mathbf{r}_t \in \mathbb{R}^{4 \times d}$ are the four region tokens at frame t (with region embeddings added), $\text{MHA}(\cdot)$ denotes standard multi-head self-attention, and $\text{LayerNorm}(\cdot)$ denotes layer normalization. The four attended region vectors are then mean-pooled to a single per-frame vector, so that, e.g., an inconsistency localized to the lips can influence the representation of the whole face region and vice versa.

E. Temporal Modeling

The per-frame fused region vector is passed through a single-layer bidirectional LSTM [?], [?] (hidden size 128 per direction, concatenated to $d = 256$) to capture frame-to-frame dynamics such as unnatural blinking or micro-expression discontinuities. The audio branch is treated symmetrically: the projected $384 \rightarrow 256$ Whisper features pass through their own single-layer BiLSTM. We deliberately use a single BiLSTM layer per branch rather than a deeper stack, since a two-layer BiLSTM was found to overfit given the size of the training data.

F. Fusion Transformer and Gated Multimodal Unit

Learned modality embeddings are added to the visual and audio token sequences, which are then concatenated along the time axis (16 tokens total: 8 visual + 8 audio) and passed through a two-layer Transformer encoder (4 heads, feedforward dimension 512, dropout 0.2) [?], allowing every visual frame to attend to every audio frame and vice versa. The fused visual and audio token streams are then separately mean-pooled back to single vectors $\mathbf{v}, \mathbf{a} \in \mathbb{R}^{256}$, and combined with a Gated Multimodal Unit [?]:

$$z = \sigma(W_g[\mathbf{v}; \mathbf{a}] + b_g), \quad (3)$$

$$\mathbf{f} = z \odot \mathbf{v} + (1 - z) \odot \mathbf{a}, \quad (4)$$

where W_g and b_g are learnable gating parameters, $\sigma(\cdot)$ is the sigmoid function, and $\odot$ denotes element-wise multiplication; $z \in (0, 1)^{256}$ is the resulting element-wise gate. Its mean over dimensions gives a single interpretable scalar in $[0, 1]$ (1 = fully visual, 0 = fully audio) reported alongside every prediction, letting the system communicate *which modality it relied on* for a given decision, rather than returning a single opaque probability.

G. Metadata, Classification, and Auxiliary Heads

A 34-dimensional container/codec feature vector – derived via FFmpeg’s `ffprobe` utility and comprising 13 normalized numeric fields (file size, duration, resolution, frame rate, bitrates, sample rate, channel count, Group of Pictures (GOP) length, frame count, rotation), one-hot video codec (5), audio codec (5), pixel format (3), and container type (3), plus 5 boolean flags (variable frame rate, missing streams, etc.) – is embedded through a small MLP ($34 \rightarrow 32$, ReLU) and concatenated with the fused representation $\mathbf{f}$. A classifier head ($\text{Linear}(288, 128) \rightarrow \text{ReLU} \rightarrow \text{Dropout}(0.3) \rightarrow \text{Linear}(128, 1)$) produces the real/fake logit. In parallel, an auxiliary linear head predicts one of 9 manipulation-type classes (real, faceswap, faceswap-wav2lip, fsgan, fsgan-wav2lip, rtvc, wav2lip, video_retalking, unknown_fake – a taxonomy unified across FakeAVCeleb’s method labels and PolyGlottFake’s sync-technique labels) directly from $\mathbf{f}$, which acts as an auxiliary loss that pushes the shared representation to be sensitive to *which kind* of artifact is present, not only to the binary real/fake decision.

H. Two-Stage Training: Frozen Backbone, then LoRA

Training proceeds in two stages. **Stage A** freezes DINOv2 and Whisper entirely and caches their outputs to disk, training only the region/audio projections, cross-region attention, BiL-STMs, fusion Transformer, GMU, and classification/auxiliary heads. **Stage B** re-opens a small number of backbone layers via LoRA [?], using the PEFT library: for DINOv2, LoRA adapters (rank $r = 8$, $\alpha = 16$, dropout 0.1) are injected into the attention QKV/output projections and both MLP layers of the last 4 of 12 ViT blocks; for Whisper’s encoder, the same LoRA configuration is injected into the last 1 of 4 encoder blocks. This targets the layers most likely to encode task-relevant, rather than purely general-purpose, visual/acoustic structure, while leaving the bulk of each pretrained backbone untouched – a deliberate compromise between adapting to the forgery-detection task and preserving the backbones’ generic representational quality (relevant to the generalization question in Section V).

I. Explainability: Grad-CAM Adaptation and Attention Rollout

Standard Grad-CAM [?] cannot be applied to DINOv2’s patch tokens as-is: in DINOv2’s `forward_features`, the CLS token and patch tokens are sibling slices of the same normalized output, and only the CLS token is consumed downstream by the visual branch, so patch tokens have no differentiable path to the logit – their gradient is identically None regardless of correct `retain_grad()` usage. We resolve this by manually replaying the last DINOv2 Transformer block with an explicit, non-fused softmax attention computation (rather than `scaled_dot_product_attention`), which exposes a real, differentiable CLS-to-patch attention row that *is* on the gradient path (the CLS output is literally an attention-weighted sum of patch values). Grad-CAM’s $\text{weight} \times \text{activation}$ formulation is then applied to that attention row instead of to patch activations directly. Separately, we apply attention rollout [?] across the two-layer fusion Transformer, folding each layer’s attention with the residual identity ($0.5 A + 0.5 I$, renormalized) to obtain a per-frame visual/audio importance trace. The Grad-CAM heatmaps, attention-rollout trace, and GMU gate value are composed into a single explanation per prediction.

IV. EXPERIMENTAL SETUP

A. Datasets and Identity-Disjoint Splitting

Table ?? summarizes the pooled dataset. FakeAVCeleb [?], PolyGlottFake [?], FaceForensics++ [?], and Celeb-DF-v2 [?] are combined into a single 44,561-video pool used for training, validation, and in-distribution testing. We flag a data-provenance correction here: an earlier version of this pipeline stored all files under a single FFPP source label, but a filename-convention audit (2026-08-16) found that bucket was only $\sim 23\%$ genuine FaceForensics++ (2,000 videos, 3-digit source/pair naming) – the majority ($\sim 72\%$, 6,229 videos) was actually Celeb-DF-v2 (`idN`-prefixed naming, matching

TABLE I
DATASET COMPOSITION (VIDEO COUNTS), CORRECTED SOURCE LABELS

Source	Real	Fake	Total
FakeAVCeleb [?]	500	21,066	21,566
PolyGlottFake [?]	762	13,605	14,367
FaceForensics++ [?]	1,000	1,000	2,000
Celeb-DF-v2 [?]	590	5,639	6,229
FFPP_unresolved (undetermined)	349	49	398
DFD-convention (1 file)	0	1	1
<i>Pooled (in-distribution)</i>	<i>3,201</i>	<i>41,360</i>	<i>44,561</i>
DFDC [?] (held out)	77	323	400

Celeb-DF-v2’s published 590-real/5,639-fake composition (almost exactly), plus one stray file matching the Google/Jigsaw DeepFake Detection (DFD) naming convention and 398 files of undetermined origin (FFPP_unresolved in Table ??, reported honestly rather than assigned a guessed source). We correct that mislabeling here rather than silently continuing to describe this pool as “FaceForensics++.” As Table ?? shows, this pool is heavily class-imbalanced (3,201 real vs. 41,360 fake, a ratio of roughly 12.9:1, with individual sources reaching up to 85:1); this imbalance is mitigated during training through both a positive-class-weighted loss term and balanced minibatch sampling (Section IV-B), and through the balanced-accuracy-based threshold selection of Section IV-D rather than a naively-maximized decision threshold. All four datasets are publicly released research benchmarks. The DFDC dataset [?] (400 videos: 77 real, 323 fake) is routed at organization time directly into a held-out cross-dataset manifest and is *never* seen by any training, validation, or threshold-selection step, by construction (source-based routing, not a post-hoc filter). We deliberately use only a 400-video subset of DFDC’s full public release (which spans over 100,000 clips): our purpose here is a targeted, exploratory out-of-distribution probe – does the model transfer to a genuinely different generation pipeline at all – rather than an exhaustive benchmark against DFDC’s own leaderboard, and the subset size was set by the compute and storage budget available for cross-dataset evaluation. As reported in Section V, the effect size we observe (AUC 0.996 in-distribution vs. 0.71 on this subset, replicated in direction across six independent model variants) is large and directionally consistent enough that we do not believe it is a small-sample artifact, but we treat the exact magnitude as indicative rather than definitive and flag this explicitly as a limitation in Section VI-C.

The pooled dataset is split **80/10/10 by identity**, not by video: an identity is assigned wholly to train, validation, or test, and every video containing that identity follows it, preventing identity leakage across splits – that is the intended design, and we describe below a real bug that violated it, found and fixed as part of the same reviewer-driven audit that surfaced the source-label correction above. Identity normalization is performed per-source, since none of the datasets share a common subject-ID scheme: FakeAVCeleb and PolyGlottFake identities are recovered from an embedded `idXXXXX`-style

token via regular expression; genuine FaceForensics++ identities are the numeric source-video id(s) encoded in the filename (XXX.mp4 or a manipulated pair XXX_YYY.mp4); Celeb-DF-v2 identities are its own idN tokens; and any video matching none of these (FFPP_unresolved, the single DFD-pattern file if unmatched) falls back to an MD5 hash of its source path, treated as a video-level identity surrogate with no real grouping.

A bug in the original version of this splitting step, now fixed. The original implementation assigned each raw identity *token* independently to a split, then routed a video to whichever split any of its identities belonged to. This does not guarantee disjointness when a single video names *multiple* identities – which every face-swap video in FakeAVCeleb, FaceForensics++, and Celeb-DF-v2 does (a source-face-donor identity and a target-video identity). If identity A’s own video was assigned to train but A was paired with identity B (assigned to test) in a different video, that swap video had to go to test – so A ended up with representation in both splits. We quantified this retroactively on the splits used to produce every result reported in this paper prior to this correction: of true identities with more than one video, **455/500 (91.0%) of FakeAVCeleb identities**, 488/1,000 (48.8%) of genuine FaceForensics++ identities, and 59/59 (100%) of Celeb-DF-v2 identities had videos spanning more than one of {train, val, test}. FakeAVCeleb – the dataset we had described as already correctly identity-disjoint – was in fact the worst offender; this had never been empirically verified before. The fix unions every pair of identities that co-occur in the same video (a standard union-find/connected-components construction) and splits at the resulting component level, so no identity can appear in more than one partition by construction. Component sizes are highly skewed (the ten largest of 15,280 components cover 54.8% of all videos), so we assign components via balanced greedy bin-packing (largest first, each placed into whichever split is furthest below its 80/10/10 target share) rather than a naive random split of components, which would badly miss the target proportions given that skew. This yields 35,649 training, 4,456 validation, and 4,456 in-distribution test videos (80.0%/10.0%/10.0% by video count) with zero identity leakage, verified across all 16,326 resulting identities post-fix.

Fixing this changed the actual train/val/test membership substantially: 19,874 videos (44.6% of the trainable pool) moved to a different split than the one used to produce the original in-distribution results. **We verified the consequence empirically, and completed a genuine validation-based re-selection between the two recipes, rather than leaving either as an open question.** We retrained Stage A once (shared across both Stage B recipes below; early-stopped epoch 11, best epoch 6, val AUC 0.9985) and then Stage B twice against the corrected splits: once with the codebase’s then-unconditional default (LoRA + augmentation + weight_decay=10⁻⁴, matching attempt 4 of Table ??), and once, after adding USE_AUGMENTATION/WEIGHT_DECAY toggles (the recipe knobs had become hardcoded once attempt 4 was historically adopted, with no way to reproduce attempt

TABLE II
CORRECTED-SPLIT RETRAINS OF BOTH RECIPES VS. THE ORIGINAL
LEAKY-SPLIT RUNS

Metric	Attempt 2 (old)	Attempt 4 (old)	Attempt 4 (new)	Attempt 2 (new)
Val AUC (training)	0.9976	0.9962	0.9994	0.9995
In-dist. Test AUC	–	–	0.9989	0.9984
In-dist. Test Acc.	–	~97–99%	99.01%	98.43%
DFDC AUC	0.6912	0.7105	0.6572	0.7091
DFDC Accuracy	55.50%	50.25%	36.50%	42.25%
DFDC Fake Recall	52.01%	42.11%	23.53%	30.03%

2 exactly without this change), with attempt 2’s exact recipe (no augmentation, weight_decay= 0). Attempt-4-recipe: early-stopped epoch 9, best epoch 5, val AUC 0.9994. Attempt-2-recipe: early-stopped epoch 14, best epoch 10, val AUC 0.9995. Full evaluation of both against the corrected in-distribution test split and DFDC, alongside the original (leaky-split) numbers for reference:

Re-selecting canonical on validation AUC, per Section V-A’s own methodology: attempt 2’s recipe wins on the corrected split (0.9995 vs. 0.9994) by the same narrow margin it won by on the original leaky split (0.9976 vs. 0.9962) – a consistent, reproducible answer under both splitting regimes. **Attempt 2’s recipe, retrained on the corrected splits, is therefore now the canonical, deployed configuration**, superseding the attempt-2-on-old-splits checkpoint that was canonical after Section V-A’s original fix. Unlike the original leaky-split comparison – where the validation-selected winner (attempt 2) had *worse* DFDC performance than the runner-up (attempt 4) – under the corrected split the validation-selected winner also generalizes better to DFDC (AUC 0.7091 vs. 0.6572, accuracy 42.25% vs. 36.50%), which is a coincidence of this particular re-run rather than a guaranteed property of validation-based selection, but a reassuring one. In-distribution performance across all four configurations remains in a tight 0.996–0.999 AUC band regardless of split correctness or recipe, confirming the identity-leakage bug did not substantially inflate the paper’s headline in-distribution claim. The DFDC six-attempt comparison of Table ?? itself predates and is unaffected by the splitting bug, since DFDC was never part of any split it touched; we leave Table ?? as the historical record of that investigation rather than retrofitting it with entries from a different splitting regime.

Multiple-seed variance study. A single retrain per recipe cannot distinguish a genuine recipe effect from ordinary training-stochasticity noise – a gap the reviewer explicitly raised. We added an opt-in global seed (fixing PyTorch, NumPy, and Python RNG state prior to model construction) and retrained the canonical (attempt 2, corrected-split) recipe two further times (seeds 2 and 3, in addition to the original, unseeded deployed run, treated as seed 1), evaluating each independently:

In-distribution AUC is remarkably stable across seeds (std 0.0008 on a 0.997–0.999 range). DFDC AUC is also fairly stable (std 0.012), and notably, the ~0.06 gap between attempt 2’s mean DFDC AUC (0.7156) and attempt 4’s corrected-split DFDC AUC (0.6572, a single run) is roughly *five times* the

TABLE III
ATTEMPT 2’S RECIPE (CORRECTED SPLIT), 3 INDEPENDENT SEEDS

Metric	Seed 1	Seed 2	Seed 3
In-dist. Test AUC	0.9984	0.9988	0.9972
DFDC AUC	0.7091	0.7295	0.7083
DFDC Accuracy	42.25%	44.75%	52.00%
Metric (mean $\pm$ sample std, $n = 3$)			Value
In-dist. Test AUC			0.9981 $\pm$ 0.0008
DFDC AUC			0.7156 $\pm$ 0.0120
DFDC Accuracy			46.33% $\pm$ 5.06 pp

attempt-2 seed-to-seed standard deviation – suggestive that the recipe difference found in Table ?? reflects a genuine effect rather than retrain noise, though we caution this compares a 3-seed spread for one recipe against a single run of the other, not a symmetric two-recipe variance comparison, and stop short of a formal significance claim. **DFDC accuracy at the selected operating threshold is markedly less stable than AUC** (std 5.06 percentage points, range 42.25–52.00%) – the three runs’ validation-derived thresholds were 0.99, 0.99, and 0.73 respectively, meaning the balanced-accuracy threshold-selection procedure of Section IV-D is itself seed-sensitive even when the underlying rank-based AUC is not. This reinforces the paper’s broader point (also made in Section VI-B) that AUC alone understates the practical variability of a deployed system’s accuracy at a fixed operating point, and extends it across random seeds as well as across configurations. We deployed seed 1’s checkpoint throughout this study (it is also, by chance, the median of the three on DFDC AUC) and restored it after evaluating seeds 2 and 3, rather than picking whichever seed happened to score highest – selecting a seed post-hoc on any held-out metric would reintroduce exactly the kind of selection bias Section V-A’s methodology exists to prevent.

B. Hardware and Reproducibility

Table ?? summarizes the hardware and software environment this project’s implementation plan specifies and was developed against. Random seed 42 is used for deterministic data-level operations (uniform frame-index sampling, the identity-disjoint train/val/test split assignment of Section IV-A, and Self-Blended Images augmentation-recipe sampling), so these steps are exactly reproducible; we did not additionally fix a global PyTorch/CUDA random seed for model weight initialization or training-loop stochasticity (e.g., dropout, `WeightedRandomSampler` draws), so bit-for-bit reproduction of a specific training run’s final weights is not guaranteed, though the data splits and preprocessing are.

C. Training Configuration

Stage A trains for up to 30 epochs (early stopping patience 5 on validation AUC) with Adam (learning rate 10^{-4} , no weight decay), batch size 8 (32 for validation). **Stage B** trains for up to 15 epochs (early stopping patience 4), with two Adam parameter groups: LoRA adapter parameters at learning rate 10^{-5} , and head parameters (classifier, fusion Transformer,

TABLE IV
HARDWARE AND SOFTWARE ENVIRONMENT

Component	Specification
GPU	NVIDIA RTX 4060 (8 GB VRAM)
RAM	16 GB
Python	3.10
CUDA (PyTorch wheel target)	12.1
Core frameworks	PyTorch, torchvision, torchaudio, Transformers, PEFT, timm

GMU, BiLSTMs – warm-started from the Stage A checkpoint) at learning rate 10^{-4} ; both groups use weight decay 10^{-4} in the canonical configuration (Section V-C). Stage B uses a physical batch size of 2 with 8 gradient accumulation steps (effective batch 16) and automatic mixed precision, since raw-crop loading and live backbone forward passes are far more memory- and I/O-intensive than Stage A’s cached-feature training. Both stages optimize a combined objective

$$\mathcal{L} = \mathcal{L}_{\text{BCE}} + 0.4 \mathcal{L}_{\text{aux}}, \quad (5)$$

where $\mathcal{L}_{\text{BCE}}$ is a weighted binary cross-entropy loss over the real/fake label (with a positive-class weight w_{pos} compensating for class imbalance) and $\mathcal{L}_{\text{aux}}$ is a categorical cross-entropy loss over the 9-way manipulation-type label (Section III-G); a `WeightedRandomSampler` additionally rebalances minibatches against the severe real:fake imbalance inherent to all three source datasets. We did not instrument per-run wall-clock training time systematically across every attempt in Table ??, so we do not report a precise duration for each; for a representative logged run, Stage A (operating on cached, frozen features, with no backbone forward/backward pass) completed in approximately 15 minutes, while Stage B – which performs live backbone forward and backward passes through the partially-unfrozen DINOv2 and Whisper encoders – is substantially slower per epoch and typically early-stops within 6–15 epochs depending on the configuration (Table ??).

D. Data Augmentation (Stage B)

To reduce reliance on dataset-specific compression/resolution fingerprints, Stage B applies one consistent augmentation recipe per video (not independently per frame, to avoid destroying the temporal signal the BiLSTM/fusion Transformer rely on): JPEG re-encoding at a random quality in [25, 90] (probability 0.2), downscale-then-upscale by a random factor in [0.4, 0.9] (probability 0.4), Gaussian blur with radius in [0.3, 2.0] (probability 0.3), horizontal flip (probability 0.5), and color jitter (brightness/contrast/saturation each scaled by a factor in [0.8, 1.2], always applied).

E. Evaluation Protocol and Threshold Selection

We report AUC (threshold-independent) alongside accuracy at an explicit operating threshold, since AUC alone can mask poor real-world behavior under severe class imbalance. The decision threshold is selected exclusively on the validation split by sweeping 48 candidate thresholds in [0.05, 0.99] and choosing the one that maximizes *balanced* accuracy (mean

of real-recall and fake-recall) rather than F1. This choice is itself an empirical finding: under this dataset’s severe fake:real imbalance (up to 85:1 in some sources), F1-maximization was found to select a degenerate threshold near 0.10, at which 25–45% of genuinely real videos were misclassified as fake in a held-out evaluation despite the model’s AUC being ≈ 0.997 – a stark illustration that a high AUC does not guarantee a well-calibrated operating point. Switching to balanced-accuracy maximization moved the selected threshold to 0.940 (validation balanced accuracy 0.976) and improved held-out evaluation accuracy on a held-back 48-video stratified sample from 83.3% to 95.8%.

V. RESULTS

A. Model Selection Methodology

Section IV-A’s six attempts (Table ??) include one Stage A prerequisite baseline (#1, the frozen-backbone checkpoint that Stage B always loads and fine-tunes from – not itself a deployable end configuration) and five candidate Stage B configurations (#2–#6). We select the deployed, “canonical” configuration from among the five Stage B candidates using *validation AUC only*, computed on the same in-distribution held-out split used for early stopping and threshold selection; DFDC is not consulted at this step. Ranked by validation AUC, attempt #2 (LoRA fine-tuning, no data augmentation, no weight decay) is highest at 0.9976, ahead of #5 (0.9972), #3 (0.9971), #4 (0.9962), and #6 (0.9945). Attempt #2 is therefore the canonical configuration reported and deployed throughout the remainder of this paper. We report DFDC results for all five Stage B configurations in Table ?? as a single blind post-hoc evaluation; no configuration is selected or re-selected on the basis of its DFDC score, and Table ?? should be read as five independent probes of the same underlying question rather than a search for the best DFDC number. Notably, attempt #2’s DFDC AUC (0.6912) is *not* the best of the five – it is one of the two weakest – which is itself informative: even genuinely DFDC-blind selection lands on a configuration with no better cross-dataset generalization than the others, reinforcing that none of the tested interventions materially change cross-dataset behavior in a way that validation performance can detect.

B. In-Distribution Performance

The canonical Stage B configuration (LoRA fine-tuning, no augmentation, no weight decay, selected on validation AUC as described above) reaches **AUC 0.9976** on the pooled in-distribution validation split. All five Stage B configurations we evaluated (Table ??) land in the AUC 0.994–0.998 range in-distribution, regardless of the generalization intervention applied – confirming that in-distribution performance is effectively saturated and is not the axis on which these interventions differ.

C. The Cross-Dataset Generalization Gap

On the fully held-out DFDC set, the canonical model’s AUC falls to **0.6912** – a drop of over 0.3 AUC points from

in-distribution performance. Because this estimate rests on only 400 videos (77 real, 323 fake), we report its uncertainty explicitly rather than treating the point estimate as exact: using the Hanley–McNeil normal approximation for AUC standard error [?] (which requires only the class counts and the point estimate, not the raw prediction scores), the 95% confidence interval is approximately [0.632, 0.751]; the corresponding Wilson score interval [?] for the 55.50% accuracy figure at this threshold is approximately [50.6%, 60.3%]. These intervals are computed analytically from the reported class counts and point estimates, and should be interpreted as approximate uncertainty estimates rather than exact (e.g., bootstrap) confidence intervals derived from resampling the raw per-video predictions. Both intervals nonetheless sit far below the in-distribution AUC (0.998) and its accompanying near-perfect accuracy, so the qualitative conclusion – a large, real generalization gap – is robust to this sample-size caveat, even though the exact DFDC AUC value should be read as approximate rather than precise. At its own validation-derived decision threshold, DFDC accuracy is 55.50%, with a pronounced asymmetry: 70.13% real-video accuracy but only 52.01% fake-video accuracy, i.e., the model systematically under-detects DFDC’s forgeries while recognizing its real videos somewhat more reliably.

D. Six Attempts to Close the Gap

Table ?? reports six independent attempts to improve DFDC generalization, holding architecture and data otherwise fixed except where noted. Note that #4–#6 are constructed cumulatively on top of #4’s weight-decay setting rather than independently on top of #2 or #3, so the six rows are not a clean one-factor-at-a-time ablation; we flag this as a methodological limitation in Section VI.

Attempt 2 (LoRA fine-tuning only, no augmentation or weight decay) has the highest validation AUC of the five Stage B candidates and is therefore the canonical, deployed configuration under our validation-only selection rule (Section V-A). We emphasize that this is *not* the configuration with the best DFDC score – attempts 4 and 5 both score higher on DFDC AUC – which is the point: a selection procedure that never looks at DFDC does not, and should not be expected to, recover whichever configuration happens to score best on it. **Attempt 4**, evaluated purely as one of five blind post-hoc DFDC probes, does reach the highest DFDC accuracy at its own decision threshold (50.25%, driven by strong real-video accuracy) among the five, and was the configuration previously deployed in an earlier iteration of this work before we corrected the selection methodology; we retain it in Table ?? and the discussion below for completeness, but it is no longer canonical. **Attempt 5** adds Self-Blended Images [?], which synthesize pseudo-forgeries from real videos alone by warping and color-jittering a copy of a face and blending it back onto itself through a landmark-derived, Gaussian-feathered convex-hull mask – intended to teach a generation-method-agnostic “blending-boundary” cue. SBI pseudo-fakes were generated for all 2,558 real training videos; the resulting

TABLE V
SIX ATTEMPTS TO CLOSE THE IN-DISTRIBUTION/CROSS-DATASET GAP. CONFIGURATION SELECTION USES ONLY THE IN-DIST. (VALIDATION) AUC COLUMN; DFDC COLUMNS ARE A BLIND POST-HOC EVALUATION AND DO NOT INFLUENCE WHICH CONFIGURATION IS CANONICAL.

#	Attempt	DFDC AUC	DFDC Acc	DFDC Real Acc	DFDC Fake Acc	In-dist. AUC	Status
1	Stage A, frozen backbone (Stage B prerequisite)	0.6914	52.00%	77.92%	45.82%	0.9978	prerequisite, not a candidate
2	Stage B, LoRA, no augmentation	0.6912	55.50%	70.13%	52.01%	0.9976	canonical (validation-selected)
3	Stage B + augmentation	0.6683	60.75%	66.23%	59.44%	0.9971	archived
4	Stage B + augmentation + weight_decay=1e-4	0.7105	50.25%	84.42%	42.11%	0.9962	archived (previously deployed)
5	#4 + Self-Blended Images (SBI) [?]	0.7127	42.75%	90.91%	31.27%	0.9972	archived
6	#4 + SRM frequency-domain branch [?]	0.6529	43.50%	88.31%	32.82%	0.9945	discontinued

model’s DFDC AUC (0.7127) is statistically indistinguishable from attempt 4’s, and its validation-derived threshold (an unusually high 0.970) yields *worse* practical DFDC accuracy (42.75%, with fake recall of only 31.27%) despite the near-identical AUC. **Attempt 6** adds a fixed, non-trainable SRM-style high-pass residual descriptor (four classic forensic kernels – horizontal edge, vertical edge, Laplacian, and the SRM “SQUARE 5×5” kernel [?] – pooled over a 4×4 spatial grid per region/frame, 64 dimensions) as a third input modality, with the GMU generalized from a two-way sigmoid gate to an n -way softmax gate. This attempt *regresses* DFDC AUC to 0.6529, worse than every prior attempt including the unmodified baseline, while leaving in-distribution performance essentially unchanged (AUC 0.9945) – indicating the added channel gave the model more capacity to fit training-distribution-specific artifacts without capturing anything that transfers to DFDC. Because the new checkpoint format was also incompatible with the better-performing weight-decay checkpoint, we reverted to the previous checkpoint and architecture configuration (restoring twelve source files to their prior state) rather than adopting the lower-performing configuration or maintaining a permanently-disabled code path, restoring attempt 4 as the deployed configuration at that point in the investigation (subsequently superseded by attempt 2 once we corrected the selection methodology to be validation-only, Section V-A); all intermediate checkpoints from every attempt are preserved for reproducibility.

Across all six attempts, DFDC AUC remains confined to a narrow 0.65–0.71 band with no clear upward trend, despite the interventions spanning regularization, data augmentation, an added synthetic-data strategy, and an added input modality – evidence that the bottleneck is not addressable by further tuning on the same three training-source distribution.

E. Diagnostic Analysis

To guide subsequent interventions, we ran two targeted diagnostics on attempt 4’s DFDC predictions before attempting the frequency-domain branch; at the time these diagnostics were run, attempt 4 was the deployed configuration (Section V-C). We have not re-run these two diagnostics against the now-canonical attempt 2, since attempt 2’s DFDC AUC (0.6912) and error pattern are close to attempt 4’s, and the diagnostic goal – characterizing *why* the gap exists at all, not why one specific configuration has it – is not tied to a specific configuration; re-running them on attempt 2 is listed

as follow-up work in Section VI. We did, however, run a third, quantitative diagnostic (boundary-cue ablation, below) directly against the canonical attempt 2 model, since it was designed specifically to test-rather-than-assume the Grad-CAM finding’s interpretation.

Error analysis. Resolution (1920×1080), frame rate (29.97), codec (H.264), and bitrate were identical between correctly- and incorrectly-classified videos, for both real and fake classes, ruling out a covariate-shift or technical-artifact explanation for the gap. Misclassifications were not borderline: the median predicted probability for the 214 missed fakes (of 323) was 0.0138 (classified with high confidence as real), versus 0.9991 for correctly-classified fakes; the 9 misclassified real videos had a median probability of 0.9995 (classified with high confidence as fake). This rules out threshold miscalibration as the explanation – the model is making high-confidence misclassifications, not borderline ones. The 214 missed fakes traced back to 160 distinct DFDC source identities with roughly proportional spread, ruling out a small-cluster explanation.

Grad-CAM spot-check. Applying the explainability pipeline of Section III-I to the six most confidently-wrong DFDC fakes, visual attention was consistently dominated by spatial boundary/edge cues – hairline-to-background boundaries, glasses-frame edges, jaw-to-background contours – and largely ignored intra-face skin texture, where classic GAN-style artifacts are typically expected. The GMU gate showed roughly balanced visual/audio contribution (48–50% visual weight), ruling out an “audio-dominant, ignoring visual evidence” explanation.

The Grad-CAM spot-check’s attention pattern motivated a natural hypothesis: that the model over-relies on blending-boundary sharpness as its primary visual forgery cue, and that DFDC’s manipulations don’t disturb that boundary the same way FFPP/SBI-style training fakes do. This hypothesis directly motivated the frequency-domain intervention in Attempt 6. However, attention concentrating on a region in a handful of hand-picked examples does not establish that the model’s predictions causally *depend* on that region, so we designed a targeted causal test rather than treating the qualitative reading as settled.

Boundary-cue ablation (causal test). On the canonical (attempt 2) model, across all 400 DFDC held-out videos, we compare five conditions: the original face crops (recon-

TABLE VI
BOUNDARY-CUE ABLATION ON DFDC HELD-OUT (CANONICAL ATTEMPT 2, $n = 400$)

Condition	AUC	95% CI	Δ
Original (reconstructed)	0.6911	[0.631, 0.751]	–
Boundary blurred	0.6850	[0.625, 0.745]	–0.006
Boundary masked	0.7071	[0.649, 0.765]	+0.016
Interior masked	0.5994	[0.532, 0.667]	–0.092
Random-region masked	0.6860	[0.626, 0.746]	–0.005

structed through the same region-derivation pipeline used for the interventions below, to keep all five conditions on identical footing); the face-contour boundary region (a landmark-derived convex-hull ring, reusing the same MediaPipe landmark detector as our SBI pipeline, Section V-C) Gaussian-blurred; the same boundary ring mean-fill-masked; the hull’s *interior* (skin) region mean-fill-masked; and an area-matched random-region control mean-fill-masked at a random location, to establish the effect of removing *any* similarly-sized patch. Table ?? reports AUC and an approximate 95% confidence interval (Hanley–McNeil, as in Section V-B; an independent-sample approximation over the same 400 paired videos, which is conservative rather than liberal for a paired difference) for each condition.

Interpretation. This result does *not* support the boundary-cue-reliance hypothesis. Boundary blurring and boundary masking change AUC by less than the area-matched random-region control does – there is nothing distinguishing the boundary from any other similarly-sized patch of the face crop, and all three of these conditions’ confidence intervals overlap heavily with the original’s. The only condition that moves AUC by more than its own confidence interval’s width is *interior* masking, which drops AUC by 0.092 – more than double the next-largest effect and outside the narrow ± 0.016 band the other three conditions occupy. The model’s DFDC-relevant discriminative signal depends far more on interior/skin-texture information than on the face-contour boundary. We interpret the Grad-CAM finding and this ablation together as follows: attention concentrating on boundary/edge regions in specific confidently-wrong examples describes *where the model looked* in those cases, but does not establish *what its predictions causally depend on* in aggregate – the two questions can and here do have different answers. Attempt 6’s frequency-domain branch was motivated by the boundary-cue reading; this result suggests that motivating hypothesis was likely incorrect, though it does not change Attempt 6’s own measured outcome (DFDC AUC 0.6529, the worst of the six) or the broader six-attempt conclusion that no single intervention closes the gap. We regard this as a useful negative result in its own right: a plausible, attention-based explanation for a generalization failure, when subjected to a causal test rather than accepted from visualization alone, did not hold up – a caution for explainability-driven diagnosis in this literature generally, and a concrete illustration of why quantitative ablation should accompany attention-map-based interpretation

rather than substitute for it. No formal paired significance test (e.g., DeLong’s test, or a bootstrap over paired per-video probability differences) was run to sharpen the interior-masking effect further; the independent-sample confidence intervals above are already conservative for a paired design, and the effect-size gap is large enough that we are confident in the qualitative ranking, but a paired test is reasonable follow-up work.

VI. DISCUSSION

A. Generalization Requires Data Diversity, Not Just Tuning

Six attempts spanning regularization, augmentation, an added synthetic-data strategy, and an added input modality all converge on the same 0.65–0.71 AUC band on DFDC. This is stronger evidence than any single result that the limiting factor is the diversity of generation-method families represented in the training data (FakeAVCeleb, PolyGlottFake, and FaceForensics++ span only three to four closely related families), not model capacity or training recipe. This reading is consistent with, and arguably strengthened by, the boundary-cue ablation of Section V-D: if the model’s DFDC-relevant signal depends primarily on interior/skin-texture information rather than boundary sharpness, that is exactly the region where generator-specific upsampling artifacts and compression fingerprints are expected to live, which is a training-data-diversity problem, not a spatial-cue problem a different architecture or loss term would fix. We view two paths as viable going forward: (i) adding training data spanning genuinely different generation methods (e.g., Celeb-DF [?], WildDeepfake, or DFD), which directly attacks the diagnosed root cause but requires substantial new data engineering; or (ii) reporting ~ 0.69 AUC as the observed generalization limit of this architecture/data combination – itself a legitimate and useful finding, and precisely the failure mode a held-out cross-dataset protocol is designed to surface. We deliberately stopped at six attempts rather than continuing to iterate against the DFDC result indefinitely, and, more importantly, we select the deployed configuration using validation performance alone (Section V-A) rather than by choosing whichever attempt happened to score best on DFDC: repeated tuning against a nominally “held-out” set, or retrospectively crowning whichever configuration wins on it, is itself a subtle form of overfitting to that set, which would defeat the purpose of holding it out in the first place.

B. AUC Is Not Sufficient Evidence of a Working System

The threshold-selection finding in Section IV-D generalizes beyond this project: a rank-based, threshold-independent metric such as AUC can be near-perfect (0.997) while the deployed system, at a poorly chosen operating threshold, is wrong on almost half of real inputs. Any reported deepfake-detection accuracy should be accompanied by the threshold-selection methodology and, ideally, by class-imbalance-robust selection criteria such as balanced accuracy rather than F1 [?], particularly given how imbalanced public forgery datasets typically are (up to 85:1 fake:real in our pooled data).

C. Limitations

The identity-leakage bug described in Section IV-A was, at the time we first identified it, the paper’s most significant open item: the deployed checkpoint had not been retrained against the corrected split, so we could not rule out that some of the reported in-distribution performance reflected leakage rather than genuine generalization. We closed this out by retraining *both* Stage B recipes (attempt 2’s and attempt 4’s) against the corrected split and re-selecting canonical by validation AUC, exactly as Section V-A’s methodology requires (full results in Table ??, Section IV-A). The outcome: in-distribution performance was not materially affected by the leakage (0.9962→0.9984 AUC for the now-canonical recipe), the same recipe (attempt 2’s) wins under both the old and corrected splits, and DFDC performance under the corrected split (AUC 0.7091) is in line with the original finding. We additionally closed the multiple-seed gap this section originally flagged as open: two further seeds of the canonical recipe (Section IV-A, Table ??) give DFDC AUC 0.7156 ± 0.0120 across three independent runs – tight enough that the ~ 0.06 gap to attempt 4’s corrected-split DFDC AUC (0.6572) looks like a genuine recipe effect rather than retrain noise, though this compares a 3-seed spread for one recipe against a single run of the other rather than a symmetric two-recipe variance study, so we stop short of a formal significance claim; extending the seed study to attempt 4’s recipe, and to the DFDC-side six-attempt comparison of Table ?? more generally, remains open. More practically significant: DFDC *accuracy* at the selected operating threshold varied far more across seeds (42.25–52.00%, std 5.06 points) than DFDC AUC did, because the balanced-accuracy threshold-selection step itself proved seed-sensitive – a finding we did not anticipate and that qualifies our threshold-selection claims in Section IV-D: a stable AUC does not imply a stable deployed operating point across retrains, only within one. The metadata-conditioning pathway (Section III-G) was, for the runs reported here, effectively inactive: an indexing mismatch between the metadata cache and the current raw-crop file paths, discovered during verification, meant the classifier’s metadata embedding received a zero-valued input for every sample. We confirmed this was not a source of shortcut learning – an always-zero input cannot leak label information – so it does not compromise the real/fake results reported in Section V, but it does mean the metadata-conditioning capability described architecturally has not yet been empirically exercised, and its value remains untested; we flag it here rather than presenting the head as validated. Our cross-dataset evaluation uses a single held-out dataset (DFDC), evaluated on the 400-video subset discussed in Section IV-A; a stronger generalization claim would require multiple independent held-out datasets, evaluated at larger scale, spanning further generation-method families. Additionally, attempts 4–6 in Table ?? are constructed cumulatively on top of attempt 4’s weight-decay setting rather than independently on top of attempts 2 or 3, so the six rows do not form a clean one-factor-at-a-time ablation; a factorial or additive-relative-

to-a single-baseline design would isolate each intervention’s individual contribution more cleanly and is left to future work. We also did not re-run two of the three Section V-D diagnostics (error analysis, Grad-CAM spot-check) against the now-canonical attempt 2 specifically, relying instead on attempt 4’s results (run when attempt 4 was deployed) as representative of the shared failure mode; confirming those two findings transfer to attempt 2 is a cheap, high-value follow-up. The third diagnostic – the boundary-cue causal ablation – was run directly against attempt 2 and is not subject to this caveat. Separately, that ablation’s confidence intervals are independent-sample approximations over what is actually a paired design (the same 400 videos across all five conditions); a proper paired significance test (DeLong’s test or a per-video bootstrap) would likely sharpen the interior-masking effect further and is left to future work, though we do not believe it would change the qualitative ranking given the effect-size gap involved.

VII. CONCLUSION AND FUTURE WORK

We presented a multi-modal deepfake detection system combining frozen self-supervised DINOv2 and Whisper backbones, LoRA fine-tuning, cross-region attention, BiLSTM temporal modeling, Transformer-based audio-visual fusion, and GMU-based adaptive gating, achieving AUC 0.996–0.998 on a large pooled in-distribution benchmark. More centrally, we used a strictly held-out cross-dataset evaluation (DFDC) to show that this in-distribution performance does not transfer: AUC falls to 0.65–0.71 across six independent mitigation strategies on the original splits, including that investigation’s validation-selected canonical configuration (0.69); after separately correcting an identity-leakage bug in the splitting logic (Section IV-A) and re-selecting canonical by the same validation-based methodology, the currently-deployed model’s DFDC AUC is 0.71, consistent with this range. Quantitative error analysis rules out covariate shift and threshold miscalibration as explanations; a Grad-CAM spot-check on six confidently-wrong examples initially suggested over-reliance on a spatial blending-boundary cue, but a subsequent causal ablation across all 400 held-out videos (Section V-D) contradicts that reading – masking the boundary region changes AUC no more than an area-matched random-region control does, while masking the *interior* skin region produces by far the largest drop (0.092 AUC). We report both the initial hypothesis and its quantitative refutation, rather than only the version that turned out to be wrong, since the methodological lesson – that attention-map-based diagnosis needs a causal check before being treated as an explanation – is itself a contribution of this work. We report the observed generalization gap as the paper’s central empirical finding regardless of the diagnostic outcome, and argue that future work on this system should prioritize training-data diversity across generation-method families – via datasets such as Celeb-DF, WildDeepfake, or DFD – over further architectural or regularization tuning against the same three training sources. We also release, as part of this investigation, three reusable diagnostic tools (a technical-covariate

error analyzer, a Grad-CAM-based cross-dataset spot-checker, and a boundary-cue masking ablation) intended to make future generalization attempts on this or similar systems more targeted than unguided empirical iteration.

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REFERENCES

- [1] R. Tolosana, R. Vera-Rodriguez, J. Fierrez, A. Morales, and J. Ortega-Garcia, "Deepfakes and beyond: A survey of face manipulation and fake detection," *Information Fusion*, vol. 64, pp. 131–148, 2020.
- [2] A. Rössler, D. Cozzolino, L. Verdoliva, C. Riess, J. Thies, and M. Nießner, "FaceForensics++: Learning to detect manipulated facial images," in *Proc. IEEE/CVF Int. Conf. Computer Vision (ICCV)*, 2019, pp. 1–11.
- [3] B. Dolhansky, J. Bitton, B. Pfau, J. Lu, R. Howes, M. Wang, and C. C. Ferrer, "The DeepFake Detection Challenge (DFDC) dataset," *arXiv preprint arXiv:2006.07397*, 2020.
- [4] H. Khalid, S. Tariq, M. Kim, and S. S. Woo, "FakeAVCeleb: A novel audio-video multimodal deepfake dataset," in *Proc. NeurIPS Datasets and Benchmarks Track*, 2021.
- [5] Y. Zhang *et al.*, "PolyGlottFake: A novel multilingual and multimodal deepfake dataset," in *Proc. IEEE Int. Conf. Multimedia and Expo Workshops (ICMEW)*, 2024.
- [6] D. Afchar, V. Nozick, J. Yamagishi, and I. Echizen, "MesoNet: A compact facial video forgery detection network," in *Proc. IEEE Int. Workshop on Information Forensics and Security (WIFS)*, 2018, pp. 1–7.
- [7] F. Chollet, "Xception: Deep learning with depthwise separable convolutions," in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2017, pp. 1251–1258.
- [8] A. Dosovitskiy *et al.*, "An image is worth 16x16 words: Transformers for image recognition at scale," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2021.
- [9] M. Oquab *et al.*, "DINOv2: Learning robust visual features without supervision," *arXiv preprint arXiv:2304.07193*, 2023.
- [10] A. Radford, J. W. Kim, T. Xu, G. Brockman, C. McLeavey, and I. Sutskever, "Robust speech recognition via large-scale weak supervision," in *Proc. Int. Conf. Machine Learning (ICML)*, 2023, pp. 28492–28518.
- [11] E. J. Hu *et al.*, "LoRA: Low-rank adaptation of large language models," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2022.
- [12] J. Arevalo, T. Solorio, M. Montes-y-Gómez, and F. A. González, "Gated multimodal units for information fusion," in *Proc. ICLR Workshop*, 2017.
- [13] A. Vaswani *et al.*, "Attention is all you need," in *Proc. Advances in Neural Information Processing Systems (NeurIPS)*, 2017, pp. 5998–6008.
- [14] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Computation*, vol. 9, no. 8, pp. 1735–1780, 1997.
- [15] M. Schuster and K. K. Paliwal, "Bidirectional recurrent neural networks," *IEEE Trans. Signal Processing*, vol. 45, no. 11, pp. 2673–2681, 1997.
- [16] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-CAM: Visual explanations from deep networks via gradient-based localization," in *Proc. IEEE Int. Conf. Computer Vision (ICCV)*, 2017, pp. 618–626.
- [17] S. Abnar and W. Zuidema, "Quantifying attention flow in transformers," in *Proc. Annual Meeting of the Association for Computational Linguistics (ACL)*, 2020, pp. 4190–4197.
- [18] K. Shiohara and T. Yamasaki, "Detecting deepfakes with self-blended images," in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2022, pp. 18720–18729.
- [19] J. Fridrich and J. Kodovský, "Rich models for steganalysis of digital images," *IEEE Trans. Information Forensics and Security*, vol. 7, no. 3, pp. 868–882, 2012.
- [20] Y. Li, X. Yang, P. Sun, H. Qi, and S. Lyu, "Celeb-DF: A large-scale challenging dataset for deepfake forensics," in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2020, pp. 3207–3216.
- [21] C. Lugaresi *et al.*, "MediaPipe: A framework for building perception pipelines," *arXiv preprint arXiv:1906.08172*, 2019.
- [22] K. Zhang, Z. Zhang, Z. Li, and Y. Qiao, "Joint face detection and alignment using multitask cascaded convolutional networks," *IEEE Signal Processing Letters*, vol. 23, no. 10, pp. 1499–1503, 2016.
- [23] D. Chicco and G. Jurman, "The advantages of the Matthews correlation coefficient (MCC) over F1 score and accuracy in binary classification evaluation," *BMC Genomics*, vol. 21, no. 6, 2020.
- [24] S.-Y. Wang, O. Wang, R. Zhang, A. Owens, and A. A. Efros, "CNN-generated images are surprisingly easy to spot... for now," in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2020, pp. 8695–8704.
- [25] D. Cozzolino, J. Thies, A. Rössler, C. Riess, M. Nießner, and L. Verdoliva, "ForensicTransfer: Weakly-supervised domain adaptation for forgery detection," *arXiv preprint arXiv:1812.02510*, 2018.
- [26] A. Haliassos, K. Vougioukas, S. Petridis, and M. Pantic, "Lips don't lie: A generalisable and robust approach to face forgery detection," in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2021, pp. 5039–5049.
- [27] J. Cao, C. Ma, T. Yao, S. Chen, S. Ding, and X. Yang, "End-to-end reconstruction-classification learning for face forgery detection," in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2022, pp. 4113–4122.
- [28] Y. Qian, G. Yin, L. Sheng, Z. Chen, and J. Shao, "Thinking in frequency: Face forgery detection by mining frequency-aware clues," in *Proc. European Conf. Computer Vision (ECCV)*, 2020, pp. 86–103.
- [29] Y. Zheng, J. Bao, D. Chen, M. Zeng, and F. Wen, "Exploring temporal coherence for more general video face forgery detection," in *Proc. IEEE/CVF Int. Conf. Computer Vision (ICCV)*, 2021, pp. 15044–15054.
- [30] L. Li, J. Bao, T. Zhang, H. Yang, D. Chen, F. Wen, and B. Guo, "Face X-Ray for more general face forgery detection," in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2020, pp. 5001–5010.
- [31] J. A. Hanley and B. J. McNeil, "The meaning and use of the area under a receiver operating characteristic (ROC) curve," *Radiology*, vol. 143, no. 1, pp. 29–36, 1982.
- [32] E. B. Wilson, "Probable inference, the law of succession, and statistical inference," *Journal of the American Statistical Association*, vol. 22, no. 158, pp. 209–212, 1927.